



LESSON 6.4

EXPLORING SLOPE

LESSON INTRODUCTION

MA.8.AR.3.2 Given a table, graph, or written description of a linear relationship, determine the slope.

LEARNING TARGETS

- I can define slope.
- I can determine the slope given two points.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

WORKING TOWARD

MA.912.AR.2.2 Write a linear two-variable equation to represent relationships between quantities from a graph, a written description, or a table of values within a mathematical or real-world context.

ESSENTIAL QUESTIONS

- How do you find slope of a line that does not pass through the origin?
- How are the constant of proportionality and the slope of the line related?

LESSON NARRATIVE

In this lesson, students develop an understanding of how to calculate the slope, what slope represents, and how it compares to the constant of proportionality. Students are guided from using a slope triangle to being able to calculate the distances given two ordered pairs.

COMPONENT SUMMARY

6.4.1 WARM-UP (~5 MIN.)

Convince class of whether the sign is mathematically accurate

6.4.3 COLLABORATION (10 MIN.)

Compare slope and constant of proportionality

6.4.5 YOUR TURN! (5 MIN)

Additional practice finding slope of a line

6.4.2 EXPLORATION (5 MIN.)

Explore slope as ratio of vertical to horizontal side lengths of triangles

6.4.4 GUIDED INSTRUCTION (15 MIN)

Practice finding slope of a line

6.4.6 WRAP-UP (~5 MIN.)

Identify and correct error in slope calculations

RESOURCES

GLOSSARY

Key Terms:

- Slope
- Hypotenuse

Additional Terms:

- Constant of proportionality
- Ratio

MATERIALS

- (Optional) Highlighters
- (Optional) Calculators

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may calculate the change in x over change in y.
- Students may have trouble visualizing the horizontal and vertical distance without drawing a triangle.

THINGS TO CONSIDER

- Consider allowing students to draw triangles throughout the lesson.
- Consider having students use different color highlighters to distinguish between the horizontal and vertical lengths.
- Allowing students to use calculators during the lesson will help students focus on discovery and process of finding slope.
- If time is an issue, then consider assigning Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 6.4.3 Collaboration, have students initially complete the problems independently. Then, ask students to discuss their work with a partner. Finally, debrief with the whole class. These discussions will help students gain confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.1.1 Participate in Effortful Learning

In 6.4.5 Your Turn!, students build perseverance as they complete challenging tasks. Through this effortful learning they develop conceptual understanding and procedural fluency.

DIFFERENTIATE INSTRUCTION

Students may struggle with calculating slope from the two ordered pairs. For students who need a visual representation of slope, have them continue to graph the points and draw the triangle.

6.4.1 WARM-UP: CONVINCE THE CLASS

QUESTIONING

Consider having students who finish early count the steps they take when walking from the classroom to a location they regularly walk to (i.e. the gymnasium or cafeteria). Ask them how many times they would need to make that trip to be at the top of the Washington Monument.



6.4.1 Warm-Up: Convince the Class

1. Write an explanation to convince the class whether this is a math fail or a reasonable approximation.



Answers will vary.

This is a reasonable approximation! If this person completes 21 steps 42 MORE times, they will complete an additional 882 steps. Since $882 + 21 = 903$, they would indeed climb the 897 steps to the top of the Washington Monument.

OR

This is not a reasonable approximation since 20 times 40 is 800 and they will not quite make it to the top.

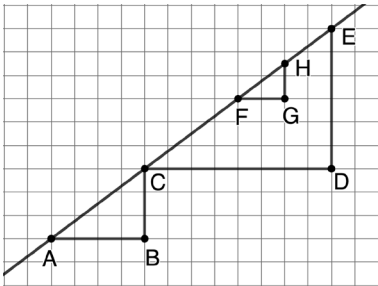


6.4.2 Exploration: Exploring Slope Through Ratios

The figure shows three right triangles, each with its **hypotenuse** on the same line. Each gridline is equal to one unit.



The **hypotenuse** is the longest side of a right triangle; the side opposite the right angle.



1. Complete the table.

Triangle	Vertical Side Length	Horizontal Side Length	$\frac{\text{vertical}}{\text{horizontal}}$	Simplified Ratio
ABC	3	4	$\frac{3}{4}$	3:4
CDE	6	8	$\frac{6}{8}$	3:4
FGH	1.5	2	$\frac{1.5}{2}$	3:4

2. What do you notice about the simplified ratios for the three triangles?
The simplified ratios are equivalent.

3. Discuss with your partner and write a conjecture about the ratio of the vertical side length to the horizontal side length of any right triangle with a hypotenuse on the same line as the triangles above.
Answers will vary. We discussed that when the right triangles have a hypotenuse on the same line, the triangles are proportional to each other. The triangles that have a hypotenuse on the same line all have equivalent ratios of the vertical length to the horizontal length.

The ratio of the vertical side length to the horizontal side length of a right triangle is known as the **slope** of the hypotenuse.



The **slope** is the ratio of the change in the vertical direction (*y*-direction) to change in the horizontal direction (*x*-direction).

6.4.2 EXPLORATION: EXPLORING SLOPE THROUGH RATIOS

TEACHER MOVES

This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group Guided Instruction. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

DIFFERENTIATE INSTRUCTION

Consider having students highlight and label the hypotenuse of each triangle in the figure to help with learning the vocabulary term.

DIFFERENTIATE INSTRUCTION

Students often mistakenly calculate the change in the horizontal direction over the change in vertical direction. Consider creating an anchor chart with instructions, visuals, and examples to help students avoid this mistake.

6.4.3 COLLABORATION: CONSTANT OF PROPORTIONALITY AND SLOPE

TEACHER MOVES

Consider summarizing and synthesizing what students learned in the Exploration before releasing them to do the Collaboration. Guide students in understanding that slope involves the ratio of the change in the dependent variables over the change in the independent variables. Keep the summary brief because the Collaboration provides further scaffolding to build a student's understanding of slope.

ENRICHMENT

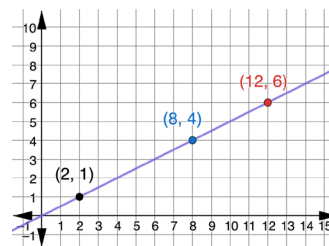
What are the similarities and differences between the slope and the constant of proportionality?

DIFFERENTIATE INSTRUCTION

Consider having students use different-colored highlighters to distinguish between the horizontal and vertical lengths.

6.4.3 Collaboration: Constant of Proportionality and Slope

1. The graph of a relationship is shown.



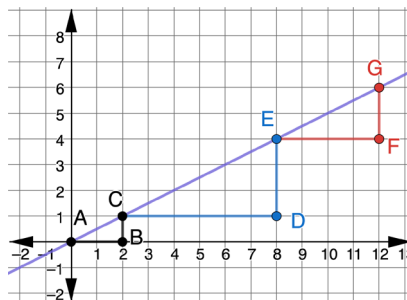
- a. Use the graph to complete the table.

x	y	Ratio $\frac{y}{x}$
2	1	$\frac{1}{2}$
4	2	$\frac{2}{4}$
8	4	$\frac{4}{8}$
10	5	$\frac{5}{10}$
12	6	$\frac{6}{12}$
16	8	$\frac{8}{16}$

- b. Use the table to complete the equation with the constant of proportionality.

$$y = \frac{1}{2}x$$

2. The graph of the same relationship is shown with three right triangles such that their hypotenuses are on the line.



- a. Complete the table for the ratio of the vertical side lengths to the horizontal side lengths of each right triangle.

Triangle	Vertical Side Length	Horizontal Side Length	$\frac{\text{vertical}}{\text{horizontal}}$
ABC	1	2	$\frac{1}{2}$
CDE	3	6	$\frac{3}{6}$
EFG	2	4	$\frac{2}{4}$

- b. What is the value of each ratio in simplest form?

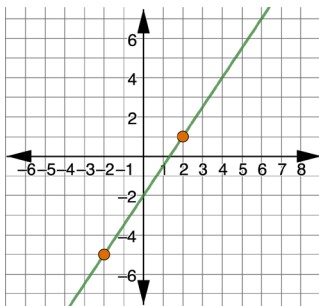
$$\frac{1}{2}$$

- c. What is the slope of the line?

$$\frac{1}{2}$$

3. Discuss with your partner how identifying the constant of proportionality is related to determining slope of a proportional relationship. Summarize your discussion.
Answers will vary. The ratio of the length of the vertical side to the horizontal side is the same as the slope of the line. The constant of proportionality is $\frac{1}{2}$ and that is the same as a slope.

The slope of any linear relationship can be determined using two points on the line.



4. Discuss with your partner how the slope of any line could be determined when right triangles are not given on a graph. Then, summarize your discussion.
Answers will vary. We could draw a right triangle on the line by using the coordinates. We can then find the ratio of the vertical side length to the horizontal side length. That will also be the slope. We could find two points on the line and calculate the ratio of the vertical change to the horizontal change.

6.4.3 COLLABORATION: CONSTANT OF PROPORTIONALITY AND SLOPE (CONTINUED)

TEACHER MOVES

Think-Pair-Share

Have students initially complete the problems independently. Then, ask students to discuss their work with a partner. Finally, debrief with the whole class. These discussions will help students gain confidence and understanding.

ENRICHMENT

- What does the slope tell us about the relationship between the independent and dependent variables?
- Is it a positive or negative relationship?
- Is there a constant or non-constant rate of change?

DIFFERENTIATE INSTRUCTION

This problem provides an opportunity to determine whether students understand the concept of slope. If students struggle to explain the concept of slope, they may benefit from continuing to draw triangles to find the slope. The triangles provide a visual representation of the rate of change. .

6.4.4 GUIDED INSTRUCTION: FINDING SLOPE GIVEN TWO POINTS

TEACHER MOVES

The Guided Instruction pulls together what the students experienced in the Exploration and the Collaboration. Use your observations from these components to determine any gaps that should be addressed during instruction. The lesson is scaffolded so that the students experience the same support as seen in the Exploration and the Collaboration. Finding slope given two points is an important process that extends into calculus. The slow development of this concept is purposeful. Students who just memorize a formula often forget it. Use the scaffolding in the lesson to build student understanding of the formula. Please note that the formula is never given as $\frac{y_2 - y_1}{x_2 - x_1}$. Most students are never sure what is meant by (x_1, y_1) , so instruction focuses on teaching students slope as the ratio of the side lengths of a slope triangle. If students begin to connect this to finding the difference between the coordinates, they may naturally develop their own formula.

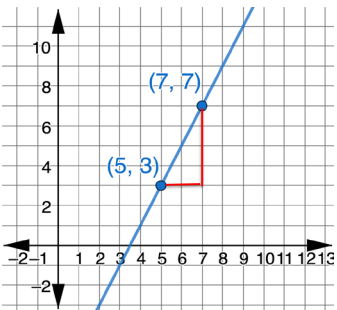
ENRICHMENT

- Why do you think the ratio for slope is found by dividing the vertical distance by the horizontal distance?
- How does this ratio connect to what you have previously learned?



6.4.4 Guided Instruction: Finding Slope Given Two Points

1. The graph of a line is shown with the coordinates of two of its points labeled.



- a. Sketch a right triangle with a hypotenuse on the line and vertices connecting the two given points.
- b. Compare your sketch to your partner's sketch.
Answer will vary. We both drew a right triangle.
- c. Use the right triangle to complete the table.

Vertical Side Length	Horizontal Side Length	$\frac{\text{vertical}}{\text{horizontal}}$	Slope
4	2	$\frac{4}{2}$	2

2. Complete the statement.
- Only one triangle

More than one triangle

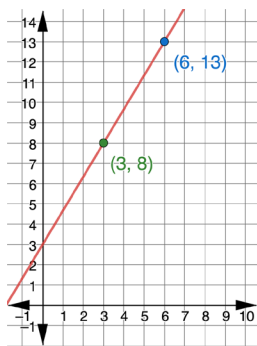
 is needed to determine the slope of the line because the slope is

always the same

sometimes different

 between any two points on the line.

3. The graph of a line is shown.
- a. Use the graph to complete the table.



Vertical Side Length	Horizontal Side Length	$\frac{\text{vertical}}{\text{horizontal}}$	Slope
5	3	$\frac{5}{3}$	$\frac{5}{3}$

- b. Without counting the units on the graph, how can the vertical distance between two points be determined?
Find the difference between the values of the y-coordinates.
- c. Without counting the units on the graph, how can the horizontal distance between two points be determined?
Find the difference between the values of the x-coordinates.

6.4.4 GUIDED INSTRUCTION: FINDING SLOPE GIVEN TWO POINTS (CONTINUED)

ENRICHMENT

How can you visually check the viability of your answers? Looking back at the slopes you have calculated in this lesson, what patterns do you notice as slope increases/ decreases?

DIFFERENTIATE INSTRUCTION

This problem may be challenging for students. Consider reviewing finding distance between two points on a coordinate plane when the points are in the same quadrant versus when the points are in different quadrants.

6.4.5 YOUR TURN!

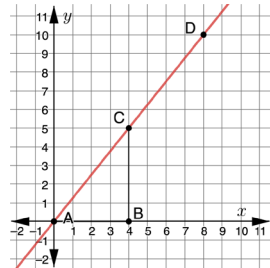
ENRICHMENT

- What is the change in x-values?
- What is the change in y-values?
- Do you find it more efficient to count the points on the graph or subtract the coordinates?



6.4.5 Your Turn!

1. A line is shown in the coordinate plane.



- a. What are the coordinates of C and D ?

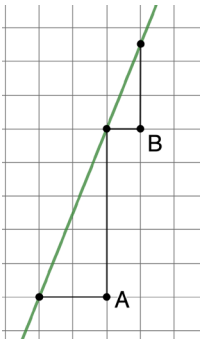
Point	Coordinates
C	$(4, 5)$
D	$(8, 10)$

- b. Determine the slope of the red line.
The slope is $\frac{5}{4}$.

TEACHER MOVES

Students might be tempted to say that Jeff is incorrect, because at first glance the slope looks different. Consider engaging students in a discussion about why we can use any two points on the line to calculate the slope, because the ratio of change in y to change in x will be the same. Connect this to students' knowledge of equivalent fractions.

2. Two triangles are shown on the graph such that their hypotenuses lie on the same green line.



Kiana uses Triangle A and says the slope of this line is $\frac{5}{2}$. Jeff uses Triangle B and argues that the slope of the green line is 2.5.

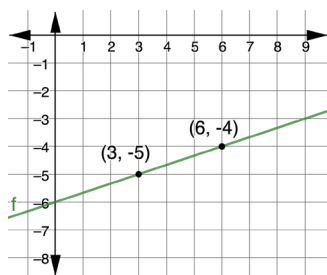
Explain who is correct.

They are both correct because Kiana's ratio of $\frac{5}{2}$ is equivalent to Jeff's ratio of $\frac{2.5}{1}$.



6.4.6 Wrap-Up: Error Analysis

1. The graph of a line is shown with the coordinates of two points labeled.



Judah found the slope of the line to be 3.

- a. Explain the error Judah made.

Answers will vary. Judah made his ratio of the vertical and horizontal change $\frac{x}{y}$ instead of $\frac{y}{x}$.

Judah should find the difference in the y-coordinates and the difference in the x-coordinates then make the ratio of $y:x$.

- b. Determine the correct slope of the line.

The slope is $\frac{1}{3}$.

6.4.6 WRAP-UP: ERROR ANALYSIS

DIFFERENTIATE INSTRUCTION

Identifying and correcting errors in samples of student work will help prevent students from making similar mistakes in their own work in the future.